Data centers and nitrate in Iowa drinking water

What the evidence supports · Policy brief · November 2026

Bottom line

- **A widely shared “zero effect” study used simulated data.** Its nitrate measurements and weather were generated by computer code, so its findings, including effects “bounded within ±0.04 mg/L,” should not be cited.
- **Real monitoring data show no detectable increase in nitrate** at the utilities that supplied, or sat near, new hyperscale campuses during 2012–2019. The data rule out effects larger than about **1 mg/L** (10% of the federal limit of 10 mg/L), but they cannot rule out smaller ones.
- **Data center land is a tiny source.** Documented central-Iowa campuses cover about 0.05% of the Raccoon River basin. Even under extreme assumptions they would add less than 0.2 mg/L to river nitrate at the intake, compared with 15–20 mg/L during the 2025 crisis.
- **The real link is water demand and treatment capacity, not water quality.** Data centers used 2–7% of West Des Moines Water Works' water in 2025, according to the utility's general manager. When nitrate-removal capacity is the bottleneck, every user's demand counts.

Why this question matters

In June 2025 Central Iowa Water Works banned lawn watering for the first time because river nitrate (above 15 mg/L) pushed its treatment plants to capacity. Data center growth in the metro led residents to ask whether data centers were responsible. County and city boards have since debated data center moratoriums, citing water among other concerns.

What went wrong with the earlier study

- Every nitrate value in its 48-system, 2005–2026 “panel” came from a random-number generator, and the code ignores real data files even when they are supplied.
- The small effect it reported was written into the generator, so the analysis found what it had assumed.
- When we applied its own method to real EPA data, its equivalence test failed (estimate −0.16 mg/L, 90% range −0.48 to 0.16).

What real data show

We used EPA compliance samples from 845 Iowa community water systems for 2012–2019 and documented construction dates for Meta (Altoona, 2013) and two Microsoft campuses (West Des Moines, 2014 and 2018). We compared changes at exposed utilities with changes at 317 utilities far from any campus, and tested the results against hundreds of “placebo” campuses placed at random elsewhere in Iowa.

Who was exposed	Estimated change in nitrate (mg/L)	Plausible range (95%)
Utility supplying the new campus	−0.14	−0.98 to 0.76
Utilities within 5 km	−0.23	−1.28 to 0.80
Utilities within 10 km	−0.14	−1.04 to 0.74

A wider 15 km comparison showed an apparent increase (0.69 mg/L), but it came from Des Moines Water Works, whose river intakes cannot receive runoff from the Meta site. That pattern reflects the 2015 river nitrate surge, not data centers. Standard statistical tests would have wrongly found “significant” effects in up to 45% of placebo comparisons.

What the data cannot tell us yet

- The public EPA files end in 2019, so they cannot speak directly to 2025–2026 or to Apple Waukee, Microsoft's newest campuses, or the Cedar Rapids projects.
- Only a few utilities were exposed, and many tests report nitrate only as “below 1 mg/L,” which hides small changes.

Implications for decision makers

- Stop citing the simulated “zero effect” study, in either direction.
- Treat water **quantity and treatment capacity** as the core data center water issue. Ask for monthly and peak-day demand, the share of nitrate-removal capacity a project would use, and commitments to low-water or closed-loop cooling during high-nitrate periods.
- Keep the focus of nitrate policy on agricultural nitrogen, the dominant source of river nitrate.
- Make post-2019 drinking-water results, utility blend and capacity records, and data center water-use permits available in downloadable form so the 2025–2026 period can be studied properly.

Source: “Data center development and nitrate in Iowa public water supplies: a forensic audit and design-based reanalysis” (working paper draft, 3-claude). Data: EPA Six-Year Review 4 (2012–2019); campus dates from company announcements and trade press; river flows from USGS. Code and data: github.com/jsemprini/datacenter-nitrate-iowa. Prepared with Claude (Anthropic); authors should verify before circulation.